Case Study 1: **Integrating Legacy Systems with Modern Cloud Infrastructure**

Problem Statement

A large financial institution has been operating for decades, relying on a complex network of legacy systems written in COBOL and Fortran. These systems handle core banking functions, such as account management, transactions, and risk assessment. The institution aims to modernize its operations by migrating to a cloud-based infrastructure and incorporating advanced analytics capabilities. However, they face significant challenges in integrating their legacy systems with modern cloud technologies while ensuring data consistency and security.

This case study highlights the complexities of integrative programming in a real-world scenario. Answer the following questions:

1. List and discuss the key challenges.

- The first key challenge that is present in this case study is the migration from COBOL and Fortran to a cloud-based infrastructure. The differences in programming languages, data formats, and protocols present a significant challenge.
- The second is data consistency. Migrating from complex programming languages to modernize cloud-based infrastructure cannot guarantee data consistency because ensuring consistent data across systems during migration is crucial. Lack of compatibility can lead to errors, and the worst case is that it can lead to financial losses.
- Lastly, it poses security risks. Integrating legacy systems with modern cloud technologies may create security vulnerabilities. Legacy systems may have different security models than modern cloud platforms. Mismatched security policies can lead to vulnerabilities.

2. List and discuss the potential solutions for this.
    - Use APIs (Application Programming Interfaces) to connect legacy systems with cloud services. APIs allow seamless communication and data exchange.
    - Convert data formats such as JSON, XML, and etc. using ETL (Extract, Transform, Load) tools.

3. List and discuss some security considerations.
    - Implement strong authentication mechanisms.
    - Encrypt data in transit (TLS / SSL) and at rest (storage encryption).
    - Maintain logs for auditing and traceability.

4. Cite at least ten references in APA format. Guide: https://www.scribbr.com/category/apastyle/

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